

2. This question is about the Solar System.

(a) **Complete** the following sentence. [2]

The Solar System was formed by the collapse of a cloud of ..... and  
..... .

(b) The table shows some information about planets in our Solar System.

|                                                | Mercury | Venus | Earth | Mars  | Jupiter | Saturn | Uranus | Neptune |
|------------------------------------------------|---------|-------|-------|-------|---------|--------|--------|---------|
| mean distance from Sun (AU)                    | 0.4     | 0.7   | 1     | 1.5   | 5.2     | 9.5    | 19.2   | 30      |
| mass compared with Earth                       | 0.055   | 0.815 | 1     | 0.107 | 318     | 95     | 15     | 17      |
| orbital time (years)                           | 0.24    | 0.60  | 1     | 2     | 12      | 30     | 84     | 160     |
| mean temperature at surface on sunny side (°C) | 430     | 465   | 20    | −20   | −150    | −170   | −200   | −210    |
| number of moons                                | 0       | 0     | 1     | 2     | 63      | 60     | 27     | 13      |

Use the information in the table to answer the questions that follow.

(i) Describe how the temperature on a planet depends on the distance from the Sun. [1]

.....  
.....

(ii) Identify the planet that does not follow this trend. [1]

.....



(iii) Pallas is an asteroid found in the asteroid belt.

I. Estimate its temperature ..... °C [1]

II. Estimate its orbital time ..... years [1]

(iv) It is suggested that the greater the mass of a planet, the more moons that will orbit around it. Explain whether the data agrees with this suggestion. [2]

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